

Multionics profiling of colorectal cancer

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1 Introduction

Colorectal cancer is the third most common cancer in the world, which also has some complex microbial features such as consistent biofilm formation in proximal but not distal colorectal cancer (1). While traditional linear reference-based analyses have identified key drivers, they often fail to capture substantial structural variation (SV), population-specific alleles, and complex strain-level diversity in both host and microbial genomes.

2 Method Overview

- Collect metagenomic data (in progress, from studies making the data available)
- Running it through the Databiomix pipeline + statistical analysis
- Planned steps involving Pangenomes:
 - Graph Construction: Building human and microbial pangenome graphs to represent population diversity.
 - Graph-Based Genotyping: Mapping reads to graphs to refine variant discovery in cancer-related regions.
 - Resolution: Analyzing host-microbiome interactions at gene- and haplotype-level resolution, capturing SVs and strain-level variations missed by linear methods.
- Metabolomics, Proteomics, etc (depending on data availability)

3 Materials

- Currently, 16S metagenomic data related to colorectal cancer
- Hopefully a biobank

4 Expected outcomes

- Methodological Validation: Finding significant taxonomic and functional shifts in the gut microbiota of CRC patients compared to controls, validating the baseline dataset. Comparing Databiomix pipeline to QIIME2 output.
- Anticipated Findings:
 - Enhanced Variant Discovery: Finding new structural variants and complex loci in cancer-associated genes.
 - Strain-Level Resolution: Precise tracking of specific microbial strains and their functional gene content, linking them to changes in host epigenetic states.
 - Integrated Signatures: Identification of novel host-microbiome interaction signatures that correlate with clinical outcomes.
- Workflow Development: Potentially help include pangenomes in the Databiomix pipeline or try to see if more functionality could be added with a new QIIME2 plugin.

5 Conclusion

- Pangenomics would enrich the understanding of colorectal cancer in connection with the gut microbiome.
- Adding more omics layers might be useful (proteomics planned).
- Access to a more comprehensive data source instead of many individual ones would increase the quality of the results.

References

1. Drewes, J. L., White, J. R., Dejea, C. M., Fathi, P., Iyadorai, T., Vadivelu, J., Roslani, A. C., Wick, E. C., Mongodin, E. F., Loke, M. F., Thulasi, K., Gan, H. M., Goh, K. L., Chong, H. Y., Kumar, S., Wanyiri, J. W., & Sears, C. L. (2017). High-resolution bacterial 16S rRNA gene profile meta-analysis and biofilm status reveal common colorectal cancer consortia. *Npj Biofilms and Microbiomes*, 3(1), 34. <https://doi.org/10.1038/s41522-017-0040-3>